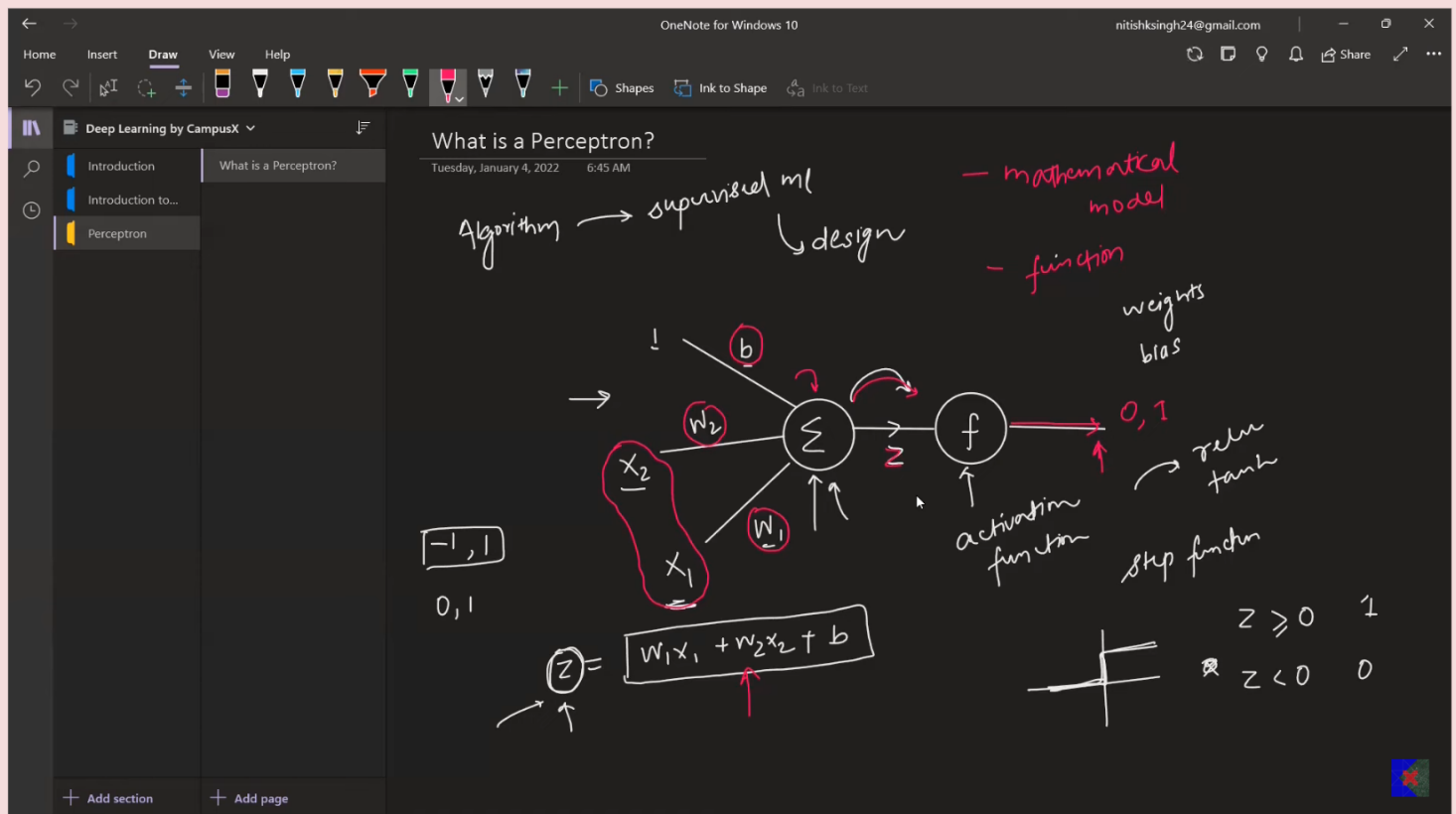


-> Perceptron is an algorithm used for supervised ml.



-> Perceptron algo diagram has input, bias and weights, summation and activation function.

-> Activation function provide us a range e.g. -1 to 1

-> step function give us the range of 0-1. 1 when $z \geq 0$ and 0 when $z < 0$

-> Using this model, what we do is:

- At first we train the model with the labeled data
- The model adjust the weight and bias base on the input data to meet the labelled requirement.
- So, in the testing phase when we enter similar data as that perticular label, it will predict that lable.
- The model wil find the sweet weight and bias for those data so that if we take input any of that data r similar data it will give us same result as label.

-> There will be n number of input if there is n-1 number of

columns or feature available in data table. 1 input would be the bias input.

What is a Perceptron? Perceptron Vs Neuron | Perceptron Geometric Intuition

Deep Learning by CampusX

Introduction

What is a Perceptron?

Introduction to...

Neuron Vs Perceptron

Tuesday, January 4, 2022 8:07 AM

billions

NEURON

PERCEPTRON

15:25 / 38:33 · Neuron Vs Perceptron >

- > Perceptron is similar as our neuron on nervous system.
- > In the model the weights tells us that how important that particular input is in relation with label. means, how well it can manipulate the label/ result.
- > so, if a input dominates the output most, in the training phase the weight will be assing high for that perticular input.

Geometric Intuition

Tuesday, January 4, 2022 8:38 AM

$z = w_1x_1 + w_2x_2 + b$

$y = f(z) = \begin{cases} 1 & z \geq 0 \\ 0 & z < 0 \end{cases}$

$w_1 \Rightarrow A \quad w_2 \Rightarrow B \quad b \Rightarrow C$

$x_1 \Rightarrow x \quad x_2 \Rightarrow y$

$Ax + By + C = 0$ line

$Ax + By + C \geq 0 \leftarrow \text{region}$

$Ax + By + C < 0 \leftarrow \text{region}$

100%

iq | cgm | plau

cgta

line

binary

classifica

- > Perceptron is nothing but a line. That divide the data into two regions
- > It is also called as binary classifier.
- > there is a drawback of this model. it only can classify linear or sort of linear data.